

Fitness Progress and Habit Tracker Database Project Report

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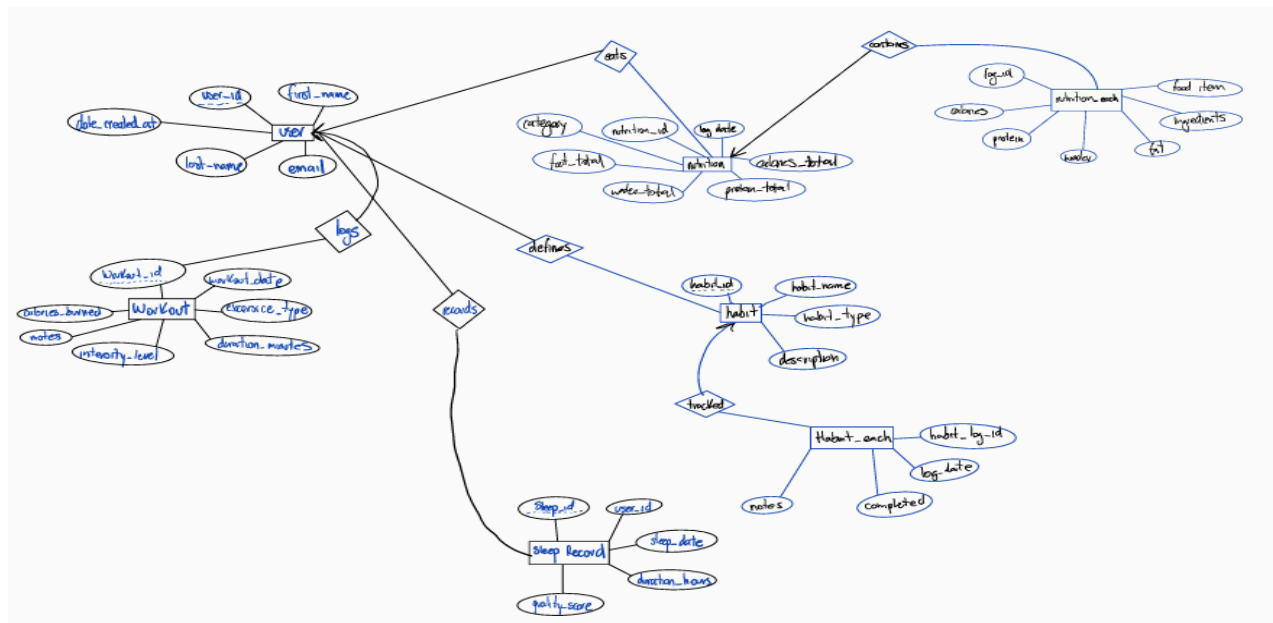
Course: COP4710

Project Title: Fitness Progress and Habit Tracker

1. Project Overview

The Fitness Progress and Habit Tracker is a database system designed to help users track their fitness, nutrition, sleep, and personal habits in one organized system. The main goal is to store health-related information in a structured database so users can review their progress over time. The system allows users to record workouts, sleep records, nutrition logs, individual food entries, habits, and habit completion logs, making it possible to track daily activity, compare workout and sleep patterns, view nutrition totals, and monitor habit progress. The database was implemented using MySQL Workbench and includes tables, relationships, constraints, sample data, queries, views, and insert, update, and delete operations.

2. ER Diagram



The ER diagram shows the main structure of the database. The central entity is the user, represented in the database as App_User. Each user can have multiple workout records, sleep

records, nutrition logs, and habits. Nutrition logs are connected to individual food entries, while habits are connected to individual habit logs.

The main relationships are:

- One user can have many workouts.
- One user can have many sleep records.
- One user can have many nutrition logs.
- One nutrition log can have many individual food entries.
- One user can have many habits.
- One habit can have many habit completion records.

These relationships help organize the data and avoid repeating the same information in multiple places.

3. Tables and Relationships

The database contains seven main tables:

App_User

The App_User table stores basic user information. Each user has a unique user_id, along with their first name, last name, email, and date created. The user_id is the primary key.

Workout

The Workout table stores exercise records for each user. It includes the workout date, exercise type, duration, calories burned, intensity level, and notes. The user_id field is a foreign key that connects each workout to a user.

Sleep_Record

The Sleep_Record table stores sleep information for each user. It includes the sleep date, number of hours slept, and sleep quality score. The user_id field connects each sleep record to the correct user.

Nutrition_Log

The Nutrition_Log table stores daily nutrition totals for each user. It includes total calories, protein, fat, and water for a specific date. Each nutrition log belongs to one user through the user_id foreign key.

Nutrition_Each

The Nutrition_Each table stores individual food items for a nutrition log. It includes the food item, category, calories, protein, fat, and water. This table is connected to Nutrition_Log using the log_id foreign key.

Habit

The Habit table stores habits that a user wants to track. Each habit has a name, type, and description. The user_id foreign key connects each habit to a user.

Habit_Each

The Habit_Each table stores individual habit log entries. It records the date, whether the habit was completed, and any notes. It connects to the Habit table using the habit_id foreign key.

4. Normalization

The database was designed to satisfy Third Normal Form.

First Normal Form

The database satisfies 1NF because all attributes are atomic, and there are no repeating groups or multi-valued fields. For example, nutrition data was separated into Nutrition_Log and Nutrition_Each so daily totals and individual food items are not stored together in one table.

Second Normal Form

The database satisfies 2NF because all tables use single-column primary keys. Since there are no composite primary keys, there are no partial dependencies. Every non-key attribute depends on the full primary key of its table.

Third Normal Form

The database satisfies 3NF because there are no transitive dependencies. Non-key attributes depend only on the primary key of their table. For example, habit details are stored in the Habit table, while individual habit completion records are stored separately in Habit_Each. This avoids repeating habit information in every habit log.

5. Implemented Operations

The database supports the required insert, update, delete, search/filter, report, and view operations.

Insert Operations

The database includes insert statements for users, workouts, sleep records, nutrition logs, food entries, habits, and habit logs. These insert statements provide sample data so the database can be tested and demonstrated.

An additional insert operation was demonstrated by adding a new food item, “Potato Chips,” to a nutrition log. After the food item was inserted, the nutrition totals were updated using a trigger.

Update Operations

The database includes update statements that modify existing records. One update changes a workout note to show that the user felt tired. Another update changes calories burned for a workout. The project also includes an update to a nutrition entry, which updates the total nutrition values in the related nutrition log.

Delete Operations

The database includes delete statements to remove data from the system. One delete operation removes a user from the App_User table. Since foreign keys use cascading deletes, related records are also removed. Another delete operation removes a food entry from Nutrition_Each, and the nutrition log totals are updated afterward.

Search and Filtering Queries

The project includes several search and filtering queries. Examples include:

- Showing all workouts for a specific user.
- Showing nutrition logs where total calories are over 500.
- Showing all habits for a specific user.
- Showing logged food for a specific user.
- Showing users with self-care habits.
- Showing users with sleep quality scores below 6.

These queries allow the database to retrieve useful information based on specific conditions.

Report Queries

The project includes report-style queries using joins, grouping, ordering, and aggregate functions. Examples include:

- Average sleep hours and sleep quality per user.
- Workout records for users whose last name ends in “n.”
- Food category counts.
- Total workout minutes and calories burned per user.
- Users who do not have any habits.

These reports help summarize the data and make the system more useful for analysis.

6. Views

The database includes three views.

Workout_Summary

The Workout_Summary view shows workout totals for each user. It includes the user's full name, total workout minutes, and average calories burned.

Workout_and_Sleep_Correlation

The Workout_and_Sleep_Correlation view compares workout data and sleep data for the same user on the same date. This can help show whether workout activity may relate to sleep duration or sleep quality.

Gained_and_Burned_Calories

The Gained_and_Burned_Calories view compares calories gained from nutrition logs with calories burned from workouts on the same date. This gives a simple way to compare calorie intake and exercise activity.

7. Challenges Faced

One challenge was organizing the final files correctly for submission, especially making sure Schema.sql included the full database structure. Another challenge was time management and working as a group because different members worked on different parts, so we had to make sure the SQL files, screenshots, report, and video all matched and were organized properly.

8. Future Improvements

If we had more time, we would improve the frontend by making the design more polished and easier to use. We would also add secure user logins, more visual charts, and extra tracking options such as weight, mood, step count, and fitness goals. These additions would make the system more complete and useful for long-term health tracking.

9. AI Usage Declaration

We used ChatGPT to assist with organizing the report, reviewing the project requirements, used Codex to help make the frontend and checking SQL/demo organization. All work was reviewed.